

Joongheon Kim

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Professor, Korea University – School of Electrical Engineering, Seoul, Korea

Visiting Professor, Seoul National University Hospital, Seoul, Korea

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Educational Backgrounds

- **University of Southern California (USC) – Viterbi School of Engineering** *Los Angeles, California, USA*
 - Ph.D. (08/2009–08/2014) in **Computer Science** (Advisor: Prof. Andreas F. Molisch, Department of Electrical Engineering) Communications, Information, Learning & Quantum (CLIQ) Group
 - M.S. (05/2014) in **Computer Science** with specialization in **High Performance Computing and Simulations**
 - M.S. (05/2012) in **Electrical Engineering**
 - **Korea University – College of Informatics** *Seoul, Republic of Korea*
 - M.S. (03/2004–02/2006) in **Computer Science and Engineering** (Advisor: Prof. Wonjun Lee)
 - B.S. (03/1999–02/2004) in **Computer Science and Engineering**
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Professional Affiliations

Full-Time

- **Korea University**, Seoul, Republic of Korea
 - Professor (03/2026–Present)**, School of Electrical Engineering (Graduate: Department of Electrical and Computer Engineering)
 - Associate Professor (03/2021–02/2026)**, School of Electrical Engineering
 - Assistant Professor (09/2019–02/2021)**, School of Electrical Engineering
 - Adjunct Professor (03/2023–Present)**, Department of Communications Engineering (Samsung Electronics)
 - Adjunct Professor (03/2021–Present)**, Department of Semiconductor Engineering (SK Hynix)
 - Adjunct Professor (11/2022–Present)**, Department of Future Science and Technology Business (Graduate School)
 - Affiliated Professor (03/2025–Present)**, Department of Defense Convergence Technology (LIG Nex1, Graduate School)
- **Chung-Ang University**, Seoul, Republic of Korea
 - Assistant Professor (03/2016–08/2019)**, School of Computer Science and Engineering
- **Intel Corporation**, Santa Clara (Silicon Valley), California, USA
 - Systems Engineer (09/2013–02/2016)**, Platform Engineering Group
- **University of Southern California – Viterbi School of Engineering**, Los Angeles, California, USA
 - Ph.D. Student in Computer Science (08/2009–08/2014)**, Thomas Lord Department of Computer Science
 - Annenberg Graduate Fellow (2009)**, 4-Year Full Scholarship for the Ph.D. Program in Computer Science
 - Research Assistant (01/2011–08/2014)**, Communications, Information, Learning & Quantum (CLIQ) Group
 - Teaching Assistant (01/2012–05/2013)**, Computer Science and Electrical Engineering Major Courses
- **InterDigital**, San Diego, California, USA
 - Intern (05/2012–08/2012)**, Wireless Systems Evolution Department
- **LG Electronics – CTO Office**, Seoul, Republic of Korea
 - Research Engineer (01/2006–08/2009)**, Multimedia Research Lab, Seocho R&D Campus

Joint Appointments for Quantum Research Collaboration

- **Seoul National University Hospital**, Seoul, Republic of Korea
 - Visiting Professor (10/2025–Present)**, Healthcare AI Research Institute

Services for National R&D

- **Presidential Council on National Artificial Intelligence Strategy**, Advisory Committee (12/2025–Present)
 - **National Research Foundation of Korea**, Review Board (11/2024–10/2026)
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University R&D/Administrative Positions

R&D Leadership Positions

- **Korea University**, Seoul, Republic of Korea
 - Director (07/2024–12/2031)**, Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center
 - Principal Investigator (07/2024–12/2031)**, Software Star-Lab (Quantum AI Empowered Second-Life Platform Technology)

Administrative Positions

- **Korea University**, Seoul, Republic of Korea
 - Vice Department Chair (01/2025–08/2025)**, Academic Affairs, School of Electrical Engineering
 - Deputy Vice President (02/2022–08/2024)**, Office of Academic Affairs
 - Dean (06/2021–08/2023)**, Center for Teaching and Learning (CTL)

Committees

- **Korea University**, Seoul, Republic of Korea
Member (12/2023–Present), Council of the Interdisciplinary Program in Quantum Information Science, Department of Physics
Member (03/2021–02/2027), Korea University 120th Anniversary Planning Committee
Member (06/2021–03/2022, 09/2022–08/2023), Academic Affair Planning Committee
Member (03/2021–08/2023), Distance Learning Management Committee
Steering Committee Member (07/2020–06/2022), Institute of Data Science (IDS)

Awards and Honors

Research and Academic Excellence (International)

- **Top Cited Article in ETRI Journal (Wiley), among work published in 2024** 2025
- **Finalist, AAAI Student Abstract and Poster Session – Oral Presentation** 2026 (Top 35), 2023 (Top 25)
- **IEEE ICTC Best Paper Award – IEEE Communications Society** 2022
- **Spotlight, Oral Presentation – ICML Workshop on Dynamic Neural Networks (2022)** 2022
- **IEEE MMTC Best Journal Paper Award – IEEE Communications Society** 2021
- **IEEE ICOIN Best Paper Award – IEEE Computer Society** 2021
- **IEEE MMTC Outstanding Young Researcher Award – IEEE Communications Society** 2020
- **IEEE Systems Journal Best Paper Award – IEEE Systems Council** 2020
- **Next Generation and Standards (NGS) Division Recognition Award – Intel Corporation** Q1/2015
- **Annenberg Graduate Fellowship Award – University of Southern California** 2009–2013
Awardeed with Ph.D. Admission: 4 Year Full Scholarship (Tuition Waiver and \$120,000 Stipend (\$30,000/year for 4 years))
- **IEEE Seoul Section Student Paper Contest** Gold (2019), Silver (2025), Bronze (2025, 2024, 2024, 2023, 2020)
- **IEEE VTS Seoul Chapter Award – IEEE Vehicular Technology Society** 2023, 2022, 2022, 2021, 2021, 2019

Research and Academic Excellence (Korea Regional, since 2016)

- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS** 2025
- **Best Paper Award, The Journal of KICS – KICS** 2024
- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS** 2023
- **Haedong Young Scholar Award – KICS and Haedong Foundation** 2018
For recognizing a researcher under the age of 40 who has made outstanding contributions to IT R&D

Korea University

- **Granite Tower Best Research Award (Top 3%), Totally, 2 times**
– 2025 (Top 3%, top 53 professors at Korea University in 2025) 05/2026
– 2024 (Top 3%, top 54 professors at Korea University in 2024) 05/2025
- **Best Research Achievement Award – Korea University, School of Electrical Engineering** 2024
- **Insung Research Grant Award – Korea University** 01/2023
For recognizing professors in top 5% research excellence during the first 3 years at Korea University
- **Granite Tower Best Teaching Award (Top 5%), Totally, 7 times**
– Probability and Random Process KECE209-02 (2025)
– Software Programming Basics GECT002-01 (2024), GECT002-06 (2025)
– Computer Language and Laboratory EGRN151-07 (2019), EGRN151-06 (2021)
– Introduction to Computers SEMI103 (2021)
– Future Mobility Technology GEQR075 (2022)
- **Best Teaching Award (Top 20%), Totally, 7 times**
– Probability and Random Process KECE209-02 (2021), KECE209-02 (2022)
– Software Programming Basics GECT002-02 (2024), GECT002-08 (2024), GECT002-09 (2025)
– Computer Language and Laboratory EGRN151-09 (2020)
– Object-Oriented Programming SEMI104 (2021)

Academic and University Services (International)

- **Certificate of Appreciation – IEEE Future Networks, AI/ML Working Group** 08/2025
- **Certificate of Appreciation – IEEE/IFIP WiOpt (2024)** 10/2024
- **Best Editor Award – ICT Express (Elsevier)** 07/2023
- **2022 Best Chapter Award, IEEE Vehicular Technology Society Chapter, Awarded as a Treasure** 2022
- **Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST)** 2021
- **Certificate of Appreciation – Department of Computer Science, University of Southern California** 2010

Academic and University Services (Korea Regional)

- **Outstanding Contribution Award**
– The Korean Institute of Communication Sciences and Information Sciences (KICS) 2025, 2024, 2021, 2019
– Korean Institute of Information Scientists and Engineers (KIISE) Information Network Society 2023, 2022
– Open Standards and ICT Association (OSIA) 2025, 2021
– The Institute of Electronics and Information Engineers (IEIE) 2025

- **Commemorative Plaque, KICS Excellent Lecture** (Title: Quantum Machine Learning and Software Technologies) 2025
- **Best Research Group Award, KICS Research Group in Quantum Communications and Quantum Computing** (Group Leader) 2025
- **Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST)** 2021

Student Awards (International)

- **IEEE TCSC Award for Excellence in Scalable Computing** (IEEE Computer Society), Awarded to *Soohyun Park* 2025
- **Young Scientist Award** (ICSANE 2025), Awarded to *Minji Lee* 2025
- **IEEE Vehicular Technology Society Student Scholarship Award (2025)**, Awarded to *Gyu Seon Kim* 2025
- **IEEE ICDCS Best Runner-Up Poster Paper Award (2025)**, Awarded to *Seok Bin Son, Soohyun Park* 2025

R&D Projects

University-Wide/Center Projects

- **Net-Zero CAFE Research Center** (07/2024–12/2031), **(Center Director)** ITRC (Korea Univ)
- **Intelligent 6G Wireless Access System Research Center** (04/2021–12/2025) 6G AI Research Center (Korea Univ)
- **K-Starlink: Dynamic Reconfigurable and Intelligent Space-Terrestrial Networks** (06/2021–05/2024) Basic Research Lab (Ajou)
- **Nano UAV Intelligence Systems Research Lab** (10/2020–08/2023) ADD Military Special Research Center (Kwangwoon)
- **5G/Unmanned Vehicle Research Center (5G/UV-RC)** (06/2020–12/2022) ITRC (Hanyang)
- **Human Resource Development for the Biomedical Unstructured Big Data Analysis** (08/2018–12/2021) ITRC (SNU-Hospital)
- **Novel Data Science Driven Framework for Efficient Network Design** (06/2017–05/2020) Basic Research Lab (Chung-Ang)
- **Intelligent Internet of Energy (IoE) Data Research Center** (02/2020–05/2020) ITRC (Kookmin)

Industry-Funded Projects

- **Advancement Technology Development for Torpedo Deception Strategies in Submarines** (11/2022–11/2026) LIG Nex1
- **Advancement Tech Dev for Submarine Target Identification and Engagement Support Intelligence** (11/2022–11/2026) LIG Nex1
- **Mapping between Real World and VR for End-Edged Cloud Real-Time VR Servers** (09/2020–11/2025) Samsung Electronics
- **Research on Learning-based Swarm Mission Planning Algorithms** (03/2024–02/2025) LIG Nex1
- **Quantum Machine Learning-based Objection Detection for Point Cloud and its Acceleration** (12/2022–04/2024) Hyundai Motors
- **Routing Algorithms for LEO Satellite Networks** (12/2022–08/2023) Solvit System
- **Optimal Positioning Algorithms for Wide-Area Relaying Networks** (12/2022–08/2023) Solvit System
- **Distributed Learning Algorithms to Build AI Models with Multi-Center Clinical Data** (11/2022–02/2023) Cipherome
- **Cellular/Wi-Fi Handover Technology Development** (02/2022–12/2022) LG Electronics CTO Division (Smart Mobility Lab)
- **Research Trends in Digital Twin Applications to Autonomous Driving** (03/2022–04/2022) Hyundai NGV
- **Distributed Learning System Design and Implementation for Clinical Applications** (02/2022–03/2022) Cipherome
- **Super-Resolution Performance Optimization in Mobile Platforms** (05/2020–08/2020) Samsung SDS
- **Deep Learning Algorithms for mVOC Concentration Analysis** (03/2020–06/2020) Samsung Electronics (C-Lab)
- **Visual Recognition Software Implementation using Deep Learning Tools** (05/2019–11/2019) Hyundai Motors
- **A Priori Techniques Research for Efficient Multi-Edge Computing** (06/2017–12/2017) Samsung Electronics (Software Center)

Government-Funded Projects

- **Quantum AI Empowered Second-Life Platform Technology** (07/2024–12/2031), **(Software Star-Lab)** IITP
- **Quantum-Empowered Spatio-Temporal Multi-Scale Digital Twin System** (03/2025–02/2028) NRF
- **6GARROW: 6G AI-Native Integrated RAN-Core Networks** (09/2024–08/2027) IITP
- **AI Bots Collaborative Platform and Self-Organizing Artificial Intelligence Technology Development** (04/2022–12/2026) IITP
- **Development of Integrated Development Framework that supports Automatic Neural Network Generation and Deployment optimized for Runtime Environment** (04/2021–12/2025) IITP
- **Quantum Hyper-Driving: Quantum-Inspired Hyper-Connected and Hyper-Sensing Autonomous Mobility** (03/2022–02/2025) NRF
- **Korea-Japan Joint Seminar Project for Generative and Multi-Modal AI Technologies** (10/2023–09/2024) NRF
- **Integrated Perception Technology Developments for Public Safety Platforms** (06/2019–05/2023) NRF
- **Development of Quantum Deep Reinforcement Learning Algorithm using QAOA** (10/2019–04/2022) NRF
- **mmWave Radar and Deep Reinforcement Learning based Optimal Policy Autonomous Driving** (06/2019–02/2022) NRF
- **Development of Privacy-Reinforcing Distributed Transfer-Iterative Learning Algorithm** (07/2019–12/2021) MHW
- **Virtual Presence in Moving Objects through 5G (PriMO-5G)** (06/2018–06/2021) IITP
- **Distributed Secure Platform for Scalable Clinical OMOP CDM Models** (04/2019–12/2020) MHW
- **mmWave High-Speed Networking Platform Design for Next-Generation Convergence Services** (06/2016–05/2019) NRF
- **Feasibility Study of 60 GHz IEEE 802.11ad for Virtual Reality (VR) Platforms** (04/2017–12/2017) IITP

Government-Funded Research Institute Projects

- **Research on Trainability and Learnability in Quantum Machine Learning Models** (03.2026–10/2027) ETRI-Affiliated Institute
- **Discovering Quantum-Advantage Cases for Industry Adoption** (09/2025–11/2025) KTL
- **LEO Satellite Routing Research using Large Language Model and Reinforcement Learning** (05/2025–11/2025) ETRI
- **Research on Generative Quantum Machine Learning Models** (04/2025–10/2025) ETRI-Affiliated Institute
- **Quantum Reinforcement Learning for Satellite Backhaul Routing in Disaster Networks** (05/2024–11/2024) ETRI
- **NOMA-based Resource Allocation Research in Space-Air-Ground Integrated Networks** (09/2023–11/2023) ETRI
- **Autonomous Intelligent COA Search Methods for Cyber-Attacks** (12/2021–11/2022) ADD
- **Research on Intelligent Agent-based CPS Security and Reliability** (04/2021–11/2021) TTA

- Multi-GPU based Automotive HPC Platform Development (04/2020–10/2020) ETRI
- Cooperative Deep Reinforcement Learning for Online Game Multi-Agents (04/2020–08/2020) ETRI
- Verification Testbed Implementation for Privacy-Preserving Trust Data Generation (10/2019–11/2019) ETRI
- Measurement and Analysis of Multi-Task GPU Scheduling Delays (05/2019–10/2019) ETRI
- Probabilistic Decision Making and Econometric Methods for Micro-Grid (05/2017–04/2019) KEPCO Research Institute
- GPU Scheduling Performance Analysis under Queueing Delay Considerations (05/2018–10/2018) ETRI
- Improving Massive Deep Learning Training via Computation and Communication Acceleration (04/2018–10/2018) ETRI
- Parsing Techniques for Artificial Neural Network (ANN) Data Processing (09/2017–11/2017) ETRI

University of Southern California, Selected Projects

- Video Aware Wireless Networks (VAWN) Research Program Intel Labs, Verizon Wireless, Cisco Systems
- 60 GHz Real-Time Wireless Video Broadcasting Disney Research Zürich
- Annenberg Graduate Fellowship Award University of Southern California

Korea University (KU), Selected Projects

- Towards Agentic AI-Enabled Edge IoT Systems for Proactive Physical World Sensing and Control (08/2025–07/2026) PKU-KU Joint Research Grant (Collaboration with Prof. Kaigui Bian at Peking Univ)
- AI Teaching Assistant Research for LLM-based Large-Scale Education (10/2024–02/2025) KU University College
- Autonomous Mobility Control using Quantum Deep Learning (03/2023–02/2024) KU Insung Research Grant
- Mobile Access Algorithm Design using Economic Theory and AI (03/2020–02/2021) KU Future Research Grant

Selected Publications

- **11747+ Citations** (H-index: 53+, i10-index: 252+), obtained from Google Scholar Profile (as of May 5, 2026)
- **Journals and Magazines: 168** publications; among them, **124** IEEE/Nature/APS publications

Books

- *Fundamentals of 6G Communications and Networking*, Springer (2024) (Editors: X. Lin, J. Zhang, Y. Liu, J. Kim)

Selected Papers

■ Top-Tier Conferences in AI (especially, Quantum AI)

- [ECCV'26] (Double Blind, Under Rebuttal)
ECCV (2026) (S. B. Son, S. Oh, J. Kim, S. Jung, S. Park)
- [KDD'26] (Under Rebuttal)
ACM KDD (2026) (H. Ahn, E. J. Roh, S. Park, W. Saad, H.-C. Lee, J. Kim)
- [ICML'26] Quantum Robust Inner Minimization for Reinforcement Learning with Quadratic Speed-Up in Query Complexity
ICML (2026) (H. K. Lee, J. Kim, S. W. Yoon) (**Acceptance Rate: 26.56%**)
- [ACL'26] Let LLM Tutors Ask First: Proactive LLM-Based Tutoring at Scale in a 1,500-Student Online Classroom
ACL (2026) (J. Lee, G. Youn, S. Lee, C. Im, J. Kim, C. Yoo)
- [ICASSP'26] How Can Quantum Deep Learning Improve Large Language Models?
IEEE ICASSP (2026), Invited (E. J. Roh, H. Ahn, S. Y.-C. Chen, S. Park, J. Kim)
- [CIKM'25] Quantum-Amplitude Embedded Adaptation for Parameter-Efficient Fine-Tuning in Large Language Models
ACM CIKM (2025) (E. J. Roh, J. Kim) (**Acceptance Rate: 30.63%**)
- [CIKM'25] Filtered One-Shot Training for Quantum Architecture Search
ACM CIKM (2025) (S. B. Son, S. Y.-C. Chen, J. Kim, S. Park) (**Acceptance Rate: 30.63%**)
- [CIKM'25] LLM-based Interactive Coding Education via Predictive Query Management and Student-Centered Fine-Tuning: Design and Implementation with 1500-Student Class Data
ACM CIKM (2025) (G. Youn, J. Lee, J. Kim, C. Yoo) (**Acceptance Rate: 30.63%**)
- [ICASSP'25] Quantum Reinforcement Learning for Coordinated Satellite Systems
IEEE ICASSP (2025), Invited (G. S. Kim, S. Chen, S. Park, J. Kim)
- [CIKM'24] Hands-On Introduction to Quantum Machine Learning
ACM CIKM (2024), Tutorial (S. Y.-C. Chen, J. Kim)
- [CIKM'23] Quantum Split Learning for Privacy-Preserving Information Management
ACM CIKM (2023) (S. Park, H. Baek, J. Kim) (**Acceptance Rate: 27.44%**)
- [CIKM'23] Logarithmic Dimension Reduction for Quantum Neural Networks
ACM CIKM (2023) (H. Baek, S. Park, J. Kim) (**Acceptance Rate: 27.44%**)
- [AAAI'23] Quantum Multi-Agent Meta Reinforcement Learning
AAAI (2023) (W. Yun, J. Park, J. Kim) (**Acceptance Rate: 19.61%**)
- [CIKM'22] Hierarchical Reinforcement Learning using Gaussian Random Trajectory Generation in Autonomous Furniture Assembly
ACM CIKM (2022) (W. Yun, D. Mohaisen, S. Jung, J.-K. Kim, J. Kim) (**Acceptance Rate: 29.64%**)
- [IJCAI'19] Randomized Adversarial Imitation Learning for Autonomous Driving
IJCAI (2019) (M. Shin, J. Kim) (**Acceptance Rate: 17.89%**)

■ Top-Tier Conferences in Networks and Systems

- [IPDPS'25] AQUA: Hardware-Agnostic Qubit Allocation for Quantum Multi-Programming
IEEE IPDPS (2025) (X. Piao, J. Shim, J. Kim, J. Kim) (**Acceptance Rate: 24.71%**)

- [WiOpt'25] *Stabilized Robust Control for Lightweight Autonomous Aircraft Mobility: A Quantum Reinforcement Learning Approach* **IEEE/IFIP WiOpt (2025)** (G. S. Kim, J. Chung, T. Q. Duong, S. Park, [J. Kim](#))
- [NOMS'25] *Joint Multi-Agent Reinforcement Learning and Message-Passing for Distributed Multi-UAV Network Management using Conflict Graphs* **IEEE/IFIP NOMS (2025)** (Y. Cho, H. Lee, S. Park, [J. Kim](#))
- [WiOpt'24] *Advanced Taxiing Path Guidance using Multi-Agent Reinforcement Learning for Air Traffic Management* **IEEE/IFIP WiOpt (2024)** (S. Lee, G. S. Kim, S. Park, [J. Kim](#))
- [WiOpt'22] *Cooperative Video Quality Adaptation for Delay-Sensitive Dynamic Streaming using Adaptive Super-Resolution* **IEEE/IFIP WiOpt (2022)** (M. Choi, W. Yun, [J. Kim](#))
- [INFOCOM'22] *Joint Superposition Coding and Training for Federated Learning over Multi-Width Neural Networks* **IEEE INFOCOM (2022)** (H. Baek, W. Yun, Y. Kwak, S. Jung, M. Ji, M. Bennis, J. Park, [J. Kim](#)) (**Acceptance Rate: 19.93%**)
- [ICDCS'20] *Understanding the Potential Risks of Sharing Elevation Information on Fitness Applications* **IEEE ICDCS (2020)** (Ü. Meteriz, N. F. Yildiran, [J. Kim](#), D. Mohaisen) (**Acceptance Rate: 17.98%**)
- [ICDCS'18] *ShmCaffe: A Distributed Deep Learning Platform with Shared Memory Buffer for HPC Architecture* **IEEE ICDCS (2018)** (S. Ahn, [J. Kim](#), E. Lim, W. Choi, A. Mohaisen, S. Kang) (**Acceptance Rate: 20.63%**)
- [ACM-MM'17] *REQUEST: Seamless Dynamic Adaptive Streaming over HTTP for Multi-Homed Smartphone under Resource Constraints* **ACM Multimedia (2017)** (J. Koo, J. Yi, [J. Kim](#), M. A. Hoque, S. Choi) (**Acceptance Rate: 27.63%**)
- [MobiSys'10] *Energy-Efficient Rate-Adaptive GPS-based Positioning for Smartphones* **ACM MobiSys (2010)** (J. Paek, [J. Kim](#), R. Govindan) (**Acceptance Rate: 19.84%**)

■ Magazines and Journals

◀ 2026 ▶

- [TVT.accepted] *Time-to-Collision Aware Autonomous Driving Motion Control under Risky Scenario Considerations*, **IEEE Transactions on Technology**, v(n):ppp-ppp (S. Kim, J. Kim, [J. Kim](#), S. Jung)
- [CM.accepted] *Sustainable 6G Non-Terrestrial Networks: Enabling Net-Zero Operation via Renewable Energy Integration and Intelligent Control*, **IEEE Communications Magazine**, v(n):ppp-ppp (H. Lee, S. Park, S. Jung, [J. Kim](#))
- [Nature.accepted] *Fourier Analysis Perspective on Quantum Neural Networks*, **Communications Physics (Nature)**, v(n):ppp-ppp (S. Oh, E. J. Roh, A. Vasilakos, S. Park, [J. Kim](#))
- [CM.accepted] *Quantum Learning for Autonomous Aircraft Control: A Reinforcement Learning Perspective*, **IEEE Communications Magazine**, v(n):ppp-ppp (G. S. Kim, J. Chung, S. Park, S. Jung, T. Q. Duong, [J. Kim](#))
- [TMC.accepted] *Joint Sustainable Control and Quantum Reinforcement Learning for Energy-Efficient Cube-Satellite Networks*, **IEEE Transactions on Mobile Computing**, 25(n):ppp-ppp (S. Park, G. S. Kim, Y. Cho, S. Jung, Z. Han, [J. Kim](#))
- [NM.accepted] *Quantum Jump to Connected Worlds: Paradigm Shifts in Mobility and Network Applications via Quantum Neural Networks*, **IEEE Network**, v(n):ppp-ppp (H. Lee, G. S. Kim, E. J. Roh, S. Jung, S. Park, T. Q. Duong, [J. Kim](#))
- [FGCS'26.09] *GraMA: A Gradient Matrix-Guided Assignment Method for Solving Qubit Mapping Problems*, **Future Generation Computer Systems**, 182:108485 (X. Piao, [J. Kim](#), J.-K. Kim)
- [EAAI'26.08] *A Comprehensive Review on Quantum Deep Neural Networks for Prognostics and Health Management: Fundamentals, Challenges and Opportunities*, **Engineering Applications of Artificial Intelligence**, 177:114991 (M. Jha, S. Chen, C. Kulkarni, [J. Kim](#))
- [IOTJ'26.06] *Quantum Federated Gradient Aggregation using the Parameter-Shift Rule*, **IEEE Internet of Things Journal**, 13(11):ppp-ppp (J. Chung, C. Im, S. Park, [J. Kim](#), W. Lee)
- [TVT'26.04] *Large-Scale Battery-Conscious Collision-Free Path-Finding for Sustainable and Autonomous Multi-AGV Mobility Control*, **IEEE Transactions on Vehicular Technology**, 75(4):5364–5375 (Y. Cho, S. Ahn, S. Park, [J. Kim](#))
- [TAES'26.03] *Quantum Reinforcement Learning for Joint Control, Communication, and Computing in Stabilized Reusable Space Rocket*, **IEEE Transactions on Aerospace and Electronic Systems**, 62:3838–3863 (G. S. Kim, J. Chung, S. Jung, S. Park, [J. Kim](#))
- [IOTJ'26.03] *Gateway-Assisted Neural Angular Routing for Hierarchical Satellite Networks*, **IEEE Internet of Things Journal**, 13(6):12322–12339 (J. Jang, I.-S. Cho, M. Shin, [J. Kim](#), S. Jung)
- [IOTJ'26.02] *Hybrid Large Language Model and Reinforcement Learning for Energy-Efficient Multisatellite Scheduling: Boosting the Performance from Scratch*, **IEEE Internet of Things Journal**, 13(3):5379–5392 (H. Ahn, G. S. Kim, I.-S. Cho, S. Jung, [J. Kim](#))
- [PRA'26.02] *Understanding Qubit Trajectories in Parametrized Quantum Circuits*, **Physical Review A**, 113:022609 (S. Oh, C. Im, S. Park, [J. Kim](#))
- [AC'26.02] *Collaborative Learning Architecture for Autonomous Excavator Planning and Execution*, **Automation in Construction**, 182:106742 (J. Cho, M. Shin, [J. Kim](#), S. Jung) (**2024 JCR Ranked 1st, in the area of Civil Engineering (1/184)**)
- [TAES.'26.01] *Integrated Control, Communication, and Computing for Mission-Critical Embedded Unmanned Aerial Vehicles*, **IEEE Transactions on Aerospace and Electronic Systems**, 62:682–703 (G. S. Kim, S. Park, S. Jung, D. Mohaisen, [J. Kim](#))
- [TMC'26.01] *Quantum Multi-Agent Reinforcement Learning for Cooperative Mobile Access in Space-Air-Ground Integrated Networks*, **IEEE Transactions on Mobile Computing**, 25(1):1200–1218 (G. S. Kim, Y. Cho, J. Chung, S. Park, S. Jung, Z. Han, [J. Kim](#))

◀ 2025 ▶

- [MM'25.10-12] *Quantum Jump to Virtual Worlds: High-Quality Multiple Virtual Meta-Space Realization in Metaverse*, **IEEE Multi-Media**, 32(4):37–44 (S. Park, [J. Kim](#))
- [TIV'25.11] *Adaptive Quantum Federated Learning for Autonomous Surveillance Multi-Drone Networks*, **IEEE Transactions on Intelligent Vehicles**, 10(11):5055–5059 (S. Park, C. Park, S. Jung, [J. Kim](#))
- [JS'25.10] *Joint Scalable Quantum Convolutional Neural Network and Reverse-Fidelity Training for High-Accurate Recognition in Unmanned Aerial Vehicle Surveillance*, **The Journal of Supercomputing**, 81(16):1495 (E. J. Roh, [J. Kim](#), S. Jung, S. Park)
- [TASE'25.09] *Adaptive Excavation Automation in Complex Soil Environments using Reinforcement Learning*, **IEEE Transactions**

- on **Automation Science and Engineering**, 22:21181–21194 (M. Shin, J. Cho, [J. Kim](#), S. Jung)
- [ESWA'25.09] Correlation-Assisted Spatio-Temporal Reinforcement Learning for Stock Revenue Maximization, *Expert Systems with Applications*, 289:128361 (J. Chung, M. Kim, S. Min, H. Choi, S. Park, [J. Kim](#))
- [CIM'25.08] Quantum-Eyes: Scalable Quantum Convolutional Neural Networks for Low-Overhead Object Detection, **IEEE Computational Intelligence Magazine**, 20(3):63–74 ([J. Kim](#), E. J. Roh, C. Im, S. Park)
- [TON'25.08] Slimmable Federated Reinforcement Learning for Energy-Efficient Proactive Caching, **IEEE Transactions on Networking**, 33(4):2079–2094 (H. Baek, G. S. Kim, S. Park, A. F. Molisch, [J. Kim](#))
- [TAES'25.08] Quantum Multiagent Reinforcement Learning for Joint Cube-Satellites and High-Altitude Long-Endurance Aerial Vehicles in SAGIN, **IEEE Transactions on Aerospace and Electronic Systems**, 61(4):9490–9510 (G. Kim, Y. Cho, S. Park, S. Jung, [J. Kim](#))
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Academic Activities and Research Supervision

Research Supervision

■ Ph.D. Alumni

- Dr. Soohyun Park (03/2019–08/2023 (PhD), 09/2023–02/2024 (Postdoc)), **Sookmyung Women's University (Professor)**
- Dr. Hankyul Baek (03/2021–02/2024 (PhD), 03/2024–03/2025 (Postdoc)), ETRI AI Safety Institute (Researcher)
- Dr. Hyunsoo Lee (03/2021–02/2026 (PhD)), LG Electronics (Researcher)
- Dr. Gyu Seon Kim (03/2023–02/2026 (PhD), 03/2026–04/2026 (Postdoc)), RealtimeVisual (Researcher, *Military Service Exception*)
- Dr. Seok Bin Son (03/2022–02/2027 (Expected) (PhD))
- Dr. Emily Jimin Roh (03/2024–02/2027 (Expected) (PhD))

■ M.S. Alumni

- Kyeongseon Kim (09/2017–08/2019), POSTECH (Ph.D. Student in Electrical Engineering)
- Dohyun Kwon (03/2018–02/2020), Hyundai Marine & Fire Insurance (Researcher)
- Dohyun Kim (03/2018–02/2020), Naver Webtoon (Researcher)
- MyungJae Shin (03/2018–02/2020), Naver (Researcher)
- Jaeho Choi (03/2019–02/2021), Korea Meteorological Administration (Researcher, *Military Service Exception*)
- Yoo Jeong (Anna) Ha (03/2021–02/2023), The University of Chicago (Ph.D. Student in Computer Science)
- Jaehyun Chung (09/2023–08/2025), Korea University (Ph.D. Student in Electrical and Computer Engineering)
- Chaemoon Im (03/2024–02/2026), Korea University Net-Zero CAFE Research Center (Researcher)
- Yeryeong Cho (03/2024–02/2026), Korea University Net-Zero CAFE Research Center (Researcher)
- Hyojun Ahn (03/2025–02/2027 (Expected))

■ Postdoctoral Scholars

- **Dr. Minseok Choi** (09/2018–02/2019, jointly with Prof. Andreas F. Molisch (USC)), **Kyung Hee University (Professor)**
- **Dr. Soyi Jung** (03/2021–08/2021, jointly with Prof. Marco Levorato (UC-Irvine)), **Ajou University (Professor)**
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IEEE Activities (Membership, Editorial Boards, and Services)

- *Senior Member* (2018–), *Member* (2006–2017)
 - *Associate Editor* (2025–), **ACM Computing Surveys**
 - *Associate Editor* (2025–), **IEEE Communications Surveys and Tutorials**
 - *Editor* (2023–), **IEEE Internet of Things Journal**
 - *Associate Editor* (2020–), **IEEE Transactions on Vehicular Technology**
 - *Guest Editor*, **Journal of Communications and Networks** (S.I. on Quantum Technologies for Communication Systems)
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